실습과제 3

```
linux@BOOK-M8LSJ7JPPD: $ source /opt/ros/jazzy/setup.bash
linux@BOOK-M8LSJ7JPPD: $ source ~/ros2_ws/install/local_setup.bash
linux@BOOK-M8LSJ7JPPD: $ cd ~/ros2_ws/src
linux@BOOK-M8LSJ7JPPD:~/ros2_ws/src$ ros2 pkg create pub2_3 --build-type ament_cmake --dependencies rclcpp geometry_msgs
going to create a new package
package name: pub2_3
destination directory: /home/linux/ros2_ws/src
package format: 3
version: 0.0.0
description: TODO: Package description
maintainer: ['linux <linux@todo.todo>']
licenses: ['TODO: License declaration']
build type: ament_cmake
dependencies: ['rclcpp', 'geometry_msgs']
creating folder ./pub2_3
creating ./pub2_3/package.xml
creating source and include folder
creating folder ./pub2_3/src
creating folder ./pub2_3/include/pub2_3
creating ./pub2_3/CMakeLists.txt

[WARNING]: Unknown license 'TODO: License declaration'.  This has been set in the package.xml, but no LICENSE file has been created.
It is recommended to use one of the ament license identifiers:
Apache-2.0
BSL-1.0
BSD-2.0
BSD-2-Clause
BSD-3-Clause
GPL-3.0-only
LGPL-3.0-only
MIT
MIT-0
linux@BOOK-M8LSJ7JPPD:~/ros2_ws/src$ tree pub2_3
pub2_3
├── CMakeLists.txt
├── include
│   └── pub2_3
├── package.xml
└── src

4 directories, 2 files
linux@BOOK-M8LSJ7JPPD:~/ros2_ws/src$ nano pub2_3/src/pub.cpp
linux@BOOK-M8LSJ7JPPD:~/ros2_ws/src$
```



```
 GNU nano 7.2                                  pub2_3/CMakeLists.txt
cmake_minimum_required(VERSION 3.16)
project(pub2_3)

# Default to C++14
if(NOT CMAKE_CXX_STANDARD)
    set(CMAKE_CXX_STANDARD 14)
endif()

if(CMAKE_COMPILER_IS_GNUCXX OR CMAKE_CXX_COMPILER_ID MATCHES "Clang")
    add_compile_options(-Wall -Wextra -Wpedantic)
endif()

# find dependencies
find_package(ament_cmake REQUIRED)
find_package(rclcpp REQUIRED)
find_package(geometry_msgs REQUIRED)

# build
add_executable(pub2_3 src/pub.cpp)
ament_target_dependencies(pub2_3 rclcpp geometry_msgs)

# Install
install(TARGETS
    pub2_3
    DESTINATION lib/${PROJECT_NAME}
)

ament_package()




                            [ Read 28 lines ]
^C Help         ^O Write Out    ^W Where Is     ^K Cut          ^T Execute      ^C Location     M-U Undo        M-A Set Mark    M-] To Bracket
^X Exit         ^R Read File    ^\ Replace      ^U Paste        ^J Justify      ^/ Go To Line   M-E Redo        M-6 Copy        ^Q Where Was
```

```
linux@BOOK-M8LSJ7JPPD:~/ros2_ws/src$ cat pub2_3/CMakeLists.txt
cmake_minimum_required(VERSION 3.16)
project(pub2_3)

# Default to C++14
if(NOT CMAKE_CXX_STANDARD)
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# find dependencies
find_package(ament_cmake REQUIRED)
find_package(rclcpp REQUIRED)
find_package(geometry_msgs REQUIRED)

# build
add_executable(pub2_3 src/pub.cpp)
ament_target_dependencies(pub2_3 rclcpp geometry_msgs)

# Install
install(TARGETS
    pub2_3
    DESTINATION lib/${PROJECT_NAME}
)

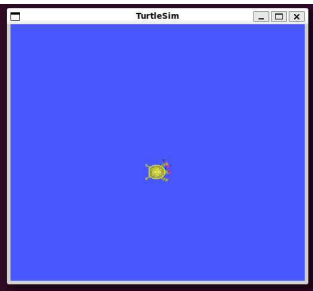
ament_package()
linux@BOOK-M8LSJ7JPPD:~/ros2_ws/src$
```

```
# build
add_executable(pub2_3 src/pub.cpp)
ament_target_dependencies(pub2_3 rclcpp geometry_msgs)

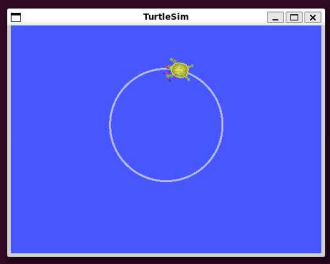
# Install
install(TARGETS
    pub2_3
    DESTINATION lib/${PROJECT_NAME}
)

ament_package()
linux@BOOK-M8LSJ7JPPD:~/ros2_ws/src$ cd ~/ros2_ws
linux@BOOK-M8LSJ7JPPD:~/ros2_ws$ colcon build --symlink-install --packages-select pub2_3
Starting >>> pub2_3
Finished <<< pub2_3 [14.2s]

Summary: 1 package finished [14.5s]
linux@BOOK-M8LSJ7JPPD:~/ros2_ws$ source install/local_setup.bash
linux@BOOK-M8LSJ7JPPD:~/ros2_ws$ ros2 run turtlesim turtlesim_node
[INFO] [1793607039.213073642] [turtlesim]: Starting turtlesim with node name /turtlesim
[INFO] [1793607039.227908873] [turtlesim]: Spawning turtle [turtle1] at x=[5.544445], y=[5.544445], theta=[0.000000]
```



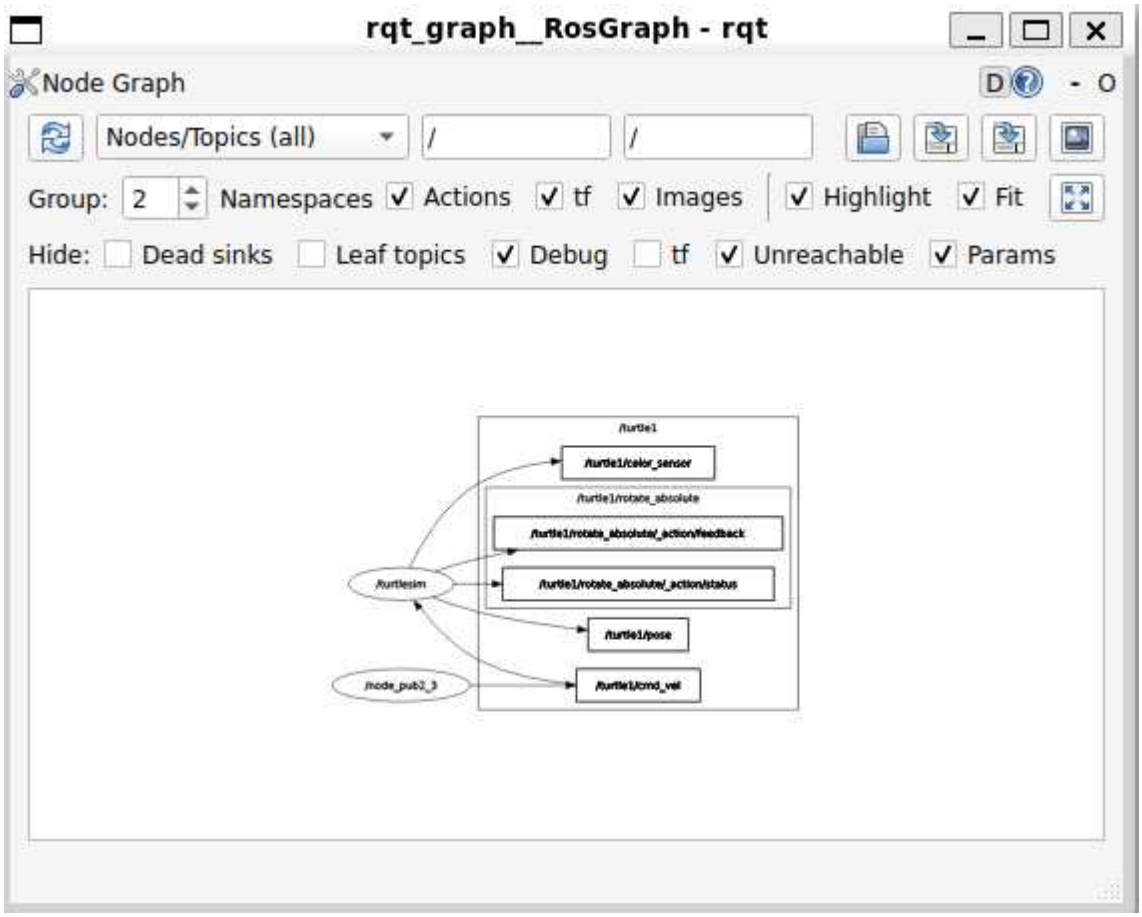

```
linux@BOOK-M8LSJ7JPPD: $ source /opt/ros/jazzy/setup.bash
linux@BOOK-M8LSJ7JPPD: $ source ~/ros2_ws/install/local_setup.bash
linux@BOOK-M8LSJ7JPPD: $ ros2 run pub2_3 pub2_3
[INFO] [1793607197.843673951] [node_pub2_3]: Publish: linear.x=2.00 angular.z=1.00
[INFO] [1793607197.943728200] [node_pub2_3]: Publish: linear.x=2.00 angular.z=1.00
[INFO] [1793607198.043689991] [node_pub2_3]: Publish: linear.x=2.00 angular.z=1.00
[INFO] [1793607198.143775741] [node_pub2_3]: Publish: linear.x=2.00 angular.z=1.00
[INFO] [1793607198.245143504] [node_pub2_3]: Publish: linear.x=2.00 angular.z=1.00
[INFO] [1793607198.343747583] [node_pub2_3]: Publish: linear.x=2.00 angular.z=1.00
[INFO] [1793607198.443822389] [node_pub2_3]: Publish: linear.x=2.00 angular.z=1.00
[INFO] [1793607198.543754235] [node_pub2_3]: Publish: linear.x=2.00 angular.z=1.00
[INFO] [1793607198.643787068] [node_pub2_3]: Publish: linear.x=2.00 angular.z=1.00
[INFO] [1793607198.743745468] [node_pub2_3]: Publish: linear.x=2.00 angular.z=1.00
[INFO] [1793607198.843730531] [node_pub2_3]: Publish: linear.x=2.00 angular.z=1.00
[INFO] [1793607198.943788743] [node_pub2_3]: Publish: linear.x=2.00 angular.z=1.00
[INFO] [1793607199.043702044] [node_pub2_3]: Publish: linear.x=2.00 angular.z=1.00
[INFO] [1793607199.143847446] [node_pub2_3]: Publish: linear.x=2.00 angular.z=1.00
[INFO] [1793607199.243753461] [node_pub2_3]: Publish: linear.x=2.00 angular.z=1.00
[INFO] [1793607199.343763912] [node_pub2_3]: Publish: linear.x=2.00 angular.z=1.00
[INFO] [1793607199.443760376] [node_pub2_3]: Publish: linear.x=2.00 angular.z=1.00
[INFO] [1793607199.546943457] [node_pub2_3]: Publish: linear.x=2.00 angular.z=1.00
[INFO] [1793607199.643726895] [node_pub2_3]: Publish: linear.x=2.00 angular.z=1.00
[INFO] [1793607199.744034243] [node_pub2_3]: Publish: linear.x=2.00 angular.z=1.00
[INFO] [1793607199.844301614] [node_pub2_3]: Publish: linear.x=2.00 angular.z=1.00
[INFO] [1793607199.943893967] [node_pub2_3]: Publish: linear.x=2.00 angular.z=1.00
[INFO] [1793607200.043930477] [node_pub2_3]: Publish: linear.x=2.00 angular.z=1.00
[INFO] [1793607200.144012390] [node_pub2_3]: Publish: linear.x=2.00 angular.z=1.00
[INFO] [1793607200.244195942] [node_pub2_3]: Publish: linear.x=2.00 angular.z=1.00
[INFO] [1793607200.343846902] [node_pub2_3]: Publish: linear.x=2.00 angular.z=1.00
[INFO] [1793607200.443783458] [node_pub2_3]: Publish: linear.x=2.00 angular.z=1.00
[INFO] [1793607200.543847893] [node_pub2_3]: Publish: linear.x=2.00 angular.z=1.00
[INFO] [1793607200.644032561] [node_pub2_3]: Publish: linear.x=2.00 angular.z=1.00
[INFO] [1793607200.743911159] [node_pub2_3]: Publish: linear.x=2.00 angular.z=1.00
[INFO] [1793607200.844377322] [node_pub2_3]: Publish: linear.x=2.00 angular.z=1.00
[INFO] [1793607200.943885904] [node_pub2_3]: Publish: linear.x=2.00 angular.z=1.00
[INFO] [1793607201.043915942] [node_pub2_3]: Publish: linear.x=2.00 angular.z=1.00
[INFO] [1793607201.143854040] [node_pub2_3]: Publish: linear.x=2.00 angular.z=1.00
[INFO] [1793607201.243842283] [node_pub2_3]: Publish: linear.x=2.00 angular.z=1.00
[INFO] [1793607201.344037868] [node_pub2_3]: Publish: linear.x=2.00 angular.z=1.00
[INFO] [1793607201.443880265] [node_pub2_3]: Publish: linear.x=2.00 angular.z=1.00
```



```
linux@BOOK-M8LSJ7JPPD:~$ source /opt/ros/jazzy/setup.bash
linux@BOOK-M8LSJ7JPPD:~$ source ~/ros2_ws/install/local_setup.bash
linux@BOOK-M8LSJ7JPPD:~$ ros2 node list
/node_pub2_3
/turtlesim
linux@BOOK-M8LSJ7JPPD:~$ ros2 topic list -t
/parameter_events [rcl_interfaces/msg/ParameterEvent]
/rosout [rcl_interfaces/msg/Log]
/turtle1/cmd_vel [geometry_msgs/msg/Twist]
/turtle1/color_sensor [turtlesim/msg/Color]
/turtle1/pose [turtlesim/msg/Pose]
linux@BOOK-M8LSJ7JPPD:~$ ros2 topic echo /turtle1/cmd_vel
linear:
  x: 2.0
  y: 0.0
  z: 0.0
angular:
  x: 0.0
  y: 0.0
  z: 1.0
---
linear:
  x: 2.0
  y: 0.0
  z: 0.0
angular:
  x: 0.0
  y: 0.0
  z: 1.0
---
linear:
  x: 2.0
  y: 0.0
  z: 0.0
angular:
  x: 0.0
  y: 0.0
  z: 1.0
---
linear:
```

```
^Clinux@BOOK-M8LSJ7JPPD:~ros2 topic hz /turtle1/cmd_velel
average rate: 10.001
        min: 0.099s max: 0.101s std dev: 0.00057s window: 12
average rate: 10.003
        min: 0.099s max: 0.101s std dev: 0.00044s window: 23
average rate: 10.000
        min: 0.099s max: 0.101s std dev: 0.00039s window: 33
average rate: 10.002
        min: 0.098s max: 0.105s std dev: 0.00094s window: 44
average rate: 10.001
        min: 0.098s max: 0.105s std dev: 0.00085s window: 55
average rate: 10.001
        min: 0.098s max: 0.105s std dev: 0.00079s window: 66
average rate: 10.001
        min: 0.098s max: 0.105s std dev: 0.00076s window: 76
average rate: 10.001
        min: 0.098s max: 0.105s std dev: 0.00071s window: 87
average rate: 10.000
        min: 0.098s max: 0.105s std dev: 0.00069s window: 97
^Clinux@BOOK-M8LSJ7JPPD:~$
```

```
^Clinux@BOOK-M8LSJ7JPPD:~ros2 topic bw /turtle1/cmd_velel
WARNING: topic [/turtle1/cmd_vel] does not appear to be published yet
Subscribed to [/turtle1/cmd_vel]
595 B/s from 6 messages
        Message size mean: 52 B min: 52 B max: 52 B
546 B/s from 16 messages
        Message size mean: 52 B min: 52 B max: 52 B
536 B/s from 26 messages
        Message size mean: 52 B min: 52 B max: 52 B
531 B/s from 36 messages
        Message size mean: 52 B min: 52 B max: 52 B
529 B/s from 46 messages
        Message size mean: 52 B min: 52 B max: 52 B
527 B/s from 56 messages
        Message size mean: 52 B min: 52 B max: 52 B
526 B/s from 66 messages
        Message size mean: 52 B min: 52 B max: 52 B
525 B/s from 76 messages
        Message size mean: 52 B min: 52 B max: 52 B
525 B/s from 86 messages
        Message size mean: 52 B min: 52 B max: 52 B
^Clinux@BOOK-M8LSJ7JPPD:~$
```


rqt_graph__RosGraph - rqt
Node Graph
Nodes/Topics (all)
/
/
Group:
2
Namespaces
Actions
tf
Images
Highlight
Fit
Hide:
Dead sinks
Leaf topics
Debug
tf
Unreachable
Params
/turtle1
/turtle1/color_sensor
/turtle1/rotate_absolute
/turtle1/rotate_absolute/_action/feedback
/turtle1/rotate_absolute/_action/status
/turtle1/pose
/turtle1/cmd_vel
/turtlesim
/node_pub2_3